Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 09

Experiment Title: Design and implement S-R, J-K & D flip flops.

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Experiment No: 09

Experiment Name:

Design and implementation of S-R, JK & D FlipFlop

Objective:

- To design JK, SR, D flipflop
- To use the clock

Apparatus:

Digital trainer board, connecting wires, IC7400, IC7404, IC7408, connecting wires.

Theory:

A flipflop is a bistable sequential circuit and are edge triggered. S-R and J-K has two inputs and a clock input. J-K is the refined version of S-R which evoids ambiguity and enables toggle D flip flop also eliminates the ambiguity.

Circuit Diagram:

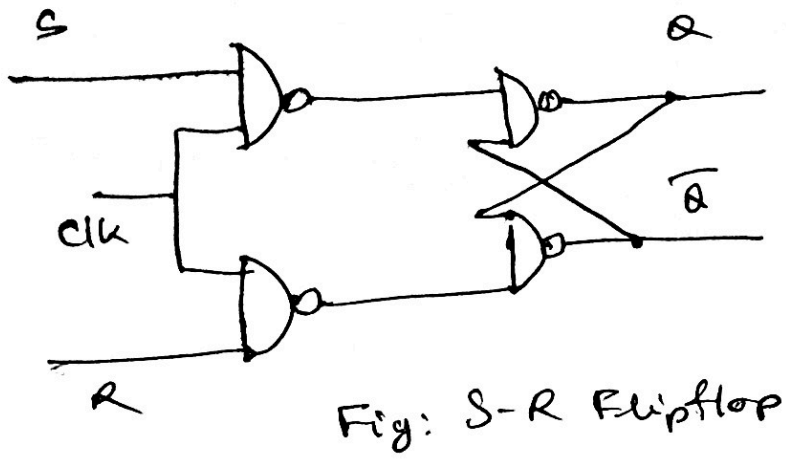


Fig: S-R Flipflop

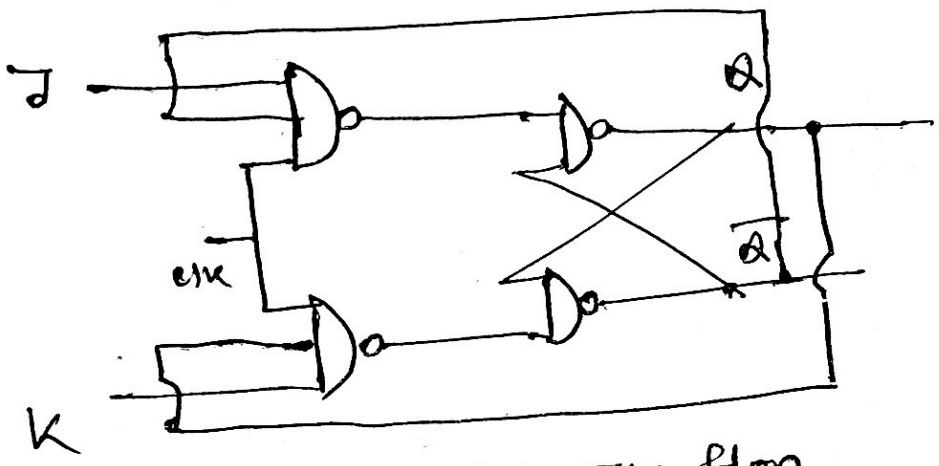


Fig: J-K Flipflop

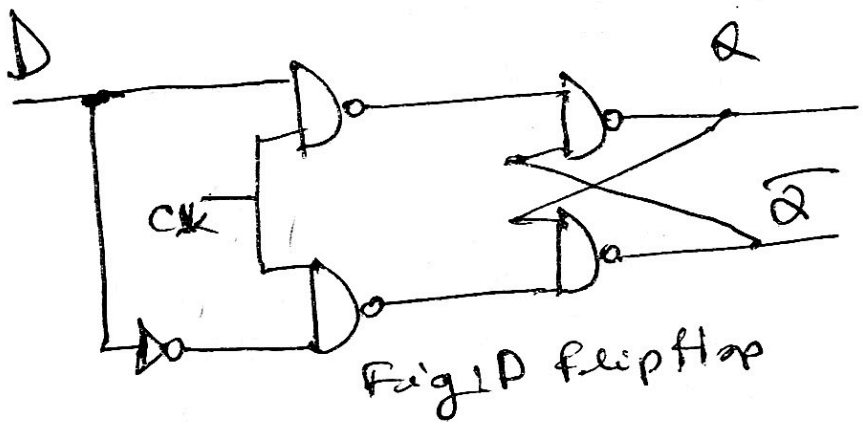
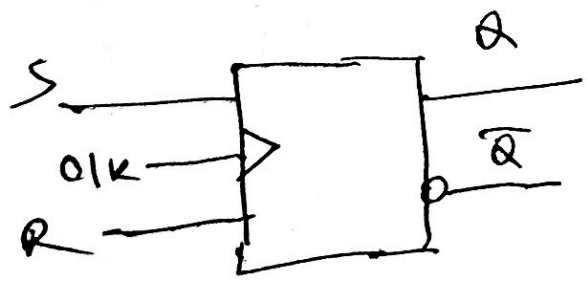
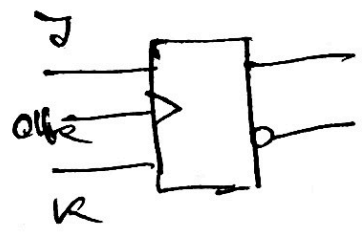
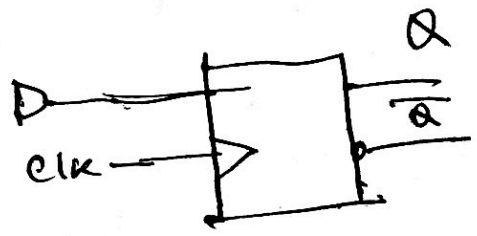


Fig: D flipflop

Block Diagram:







Truth Table:

clk	S	R	Q^{t+1}	Q^t
↓	0	0	0	0 } NC
			1	1
↓	0	1	0	0 } Reset
			0	0
↓	1	0	1	1 } Set
			1	1
↓	1	1	?	* } Inv

S-R Flipflop

clk	D	Q^{t+1}	Q^t
↓	0	0	0 } R
		0	0
↓	1	1	1 } S
		1	1

D flip flop

clk	J	K	Q^{t+1}	Q^t
↓	0	0	0	0 } NC
			1	1
↓	0	1	0	0 } R
			0	0
↓	1	0	1	1 } S
			1	1
↓	1	1	0	0 } Tog.
			1	1

J-K flip flop

Procedures:

1. Place all ICs on the bread board
2. Connect Vcc and Ground properly
3. Implement SR, JK, D flip flops
4. Apply different inputs.
5. Observe the outputs and check the truth table.

Result:

The S-R, J-K and D flipflops were successfully implemented and designed. All the outputs matched with the truth table.

Analysis:

This experiment demonstrates the working of three fundamental flip flops.

In S-R there was ambiguity.

In D- flip flop. There were only Set and Reset states.

In J-K there was no ambiguity and toggle mode was working.